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# TASK 5: EXPLORATORY DATA ANALYSIS (EDA) - TITANIC DATASET
# Tools: Python, Pandas, NumPy, Matplotlib
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# 1. IMPORT LIBRARIES
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

from google.colab import files
from IPython.display import display, Markdown

print("=" * 65)
print("TASK 5: TITANIC EXPLORATORY DATA ANALYSIS")
print("=" * 65)

# 2. UPLOAD AND LOAD DATASET
print("\nPlease upload your Titanic train.csv file.")
uploaded = files.upload()

csv_files = [
    name for name in uploaded
    if name.lower().endswith(".csv")
]

if not csv_files:
    raise ValueError("No CSV file uploaded. Please upload train.csv.")

df = pd.read_csv(csv_files[0])

print("\nDataset loaded successfully!")
print("Dataset dimensions:", df.shape)
print("\nFirst five rows:")
display(df.head())

# 3. DATASET OVERVIEW
display(Markdown("## 1. Dataset Overview"))

print("Column names:")
print(df.columns.tolist())

print("\nDataset information:")
df.info()

print("\nSummary statistics:")
display(df.describe(include="all").T)

# 4. DATA QUALITY ANALYSIS
display(Markdown("## 2. Missing Values and Duplicate Records"))

missing_summary = pd.DataFrame({
    "Missing Values": df.isnull().sum(),
    "Missing Percentage (%)":
        (df.isnull().sum() / len(df) * 100).round(2)
})
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display(
    missing_summary.sort_values(
        "Missing Values", ascending=False
    )
)

print("Total duplicate rows:", df.duplicated().sum())

# 5. SURVIVAL ANALYSIS
display(Markdown("## 3. Passenger Survival Analysis"))

if "Survived" in df.columns:
    survival_counts = df["Survived"].value_counts().reindex(
        [0, 1], fill_value=0
    )

    plt.figure(figsize=(8, 5))
    plt.bar(
        ["Did Not Survive", "Survived"],
        survival_counts.values
    )
    plt.title("Titanic Passenger Survival Counts")
    plt.xlabel("Survival Status")
    plt.ylabel("Number of Passengers")
    plt.tight_layout()
    plt.show()

    print(
        f"Overall survival rate: "
        f"{df['Survived'].mean() * 100:.2f}%"
    )
else:
    print("The dataset does not contain a Survived column.")

# 6. SURVIVAL BY SEX
display(Markdown("## 4. Survival Rate by Sex"))

if {"Sex", "Survived"}.issubset(df.columns):
    sex_survival = (
        df.groupby("Sex")["Survived"].mean() * 100
    ).sort_values(ascending=False)

    display(
        sex_survival.round(2).rename(
            "Survival Rate (%)"
        ).to_frame()
    )

    plt.figure(figsize=(8, 5))
    plt.bar(sex_survival.index, sex_survival.values)
    plt.title("Survival Rate by Sex")
    plt.xlabel("Sex")
    plt.ylabel("Survival Rate (%)")
    plt.ylim(0, 100)
    plt.tight_layout()
    plt.show()

# 7. SURVIVAL BY PASSENGER CLASS
display(Markdown("## 5. Survival Rate by Passenger Class"))
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if {"Pclass", "Survived"}.issubset(df.columns):
    class_survival = (
        df.groupby("Pclass")["Survived"].mean() * 100
    ).sort_index()

    display(
        class_survival.round(2).rename(
            "Survival Rate (%)"
        ).to_frame()
    )

    plt.figure(figsize=(8, 5))
    plt.bar(
        class_survival.index.astype(str),
        class_survival.values
    )
    plt.title("Survival Rate by Passenger Class")
    plt.xlabel("Passenger Class")
    plt.ylabel("Survival Rate (%)")
    plt.ylim(0, 100)
    plt.tight_layout()
    plt.show()

# 8. AGE DISTRIBUTION
display(Markdown("## 6. Age Distribution"))

if "Age" in df.columns:
    plt.figure(figsize=(8, 5))
    df["Age"].dropna().hist(bins=25)
    plt.title("Age Distribution of Passengers")
    plt.xlabel("Age")
    plt.ylabel("Number of Passengers")
    plt.tight_layout()
    plt.show()

    print("Missing age values:", df["Age"].isnull().sum())
    print("Median passenger age:", round(df["Age"].median(), 2))

# 9. FARE DISTRIBUTION
display(Markdown("## 7. Fare Distribution"))

if "Fare" in df.columns:
    plt.figure(figsize=(8, 5))
    df["Fare"].dropna().hist(bins=30)
    plt.title("Distribution of Passenger Fares")
    plt.xlabel("Fare")
    plt.ylabel("Number of Passengers")
    plt.tight_layout()
    plt.show()

    print("Average fare:", round(df["Fare"].mean(), 2))
    print("Median fare:", round(df["Fare"].median(), 2))

# 10. CORRELATION ANALYSIS
display(Markdown("## 8. Correlation Analysis"))

numeric_df = df.select_dtypes(include="number")

if not numeric_df.empty:
    correlation = numeric_df.corr()
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        display(correlation.round(2))

        plt.figure(figsize=(10, 7))
        plt.imshow(
            correlation,
            aspect="auto",
            vmin=-1,
            vmax=1
        )
        plt.colorbar(label="Correlation")

        plt.xticks(
            range(len(correlation.columns)),
            correlation.columns,
            rotation=45,
            ha="right"
        )
        plt.yticks(
            range(len(correlation.index)),
            correlation.index
        )

        plt.title("Correlation Matrix of Numerical Variables")
        plt.tight_layout()
        plt.show()

    # 11. ADDITIONAL INSIGHTS: SURVIVAL BY AGE GROUP
    display(Markdown("## 9. Survival by Age Group"))

    if {"Age", "Survived"}.issubset(df.columns):
        age_data = df[["Age", "Survived"]].dropna().copy()

        age_data["Age Group"] = pd.cut(
            age_data["Age"],
            bins=[0, 12, 18, 35, 60, 100],
            labels=[
                "Child (0-12)",
                "Teen (13-18)",
                "Young Adult (19-35)",
                "Adult (36-60)",
                "Senior (60+)"
            ],
            include_lowest=True
        )

        age_survival = (
            age_data.groupby(
                "Age Group", observed=False
            )["Survived"].mean() * 100
        )

        display(
            age_survival.round(2).rename(
                "Survival Rate (%)"
            ).to_frame()
        )

        plt.figure(figsize=(9, 5))
        plt.bar(
            age_survival.index.astype(str),
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            age_survival.values
        )
        plt.title("Survival Rate by Age Group")
        plt.xlabel("Age Group")
        plt.ylabel("Survival Rate (%)")
        plt.xticks(rotation=25, ha="right")
        plt.ylim(0, 100)
        plt.tight_layout()
        plt.show()

    # 12. KEY FINDINGS
    display(Markdown("## 10. Key Findings"))

    print("=" * 55)
    print("KEY FINDINGS FROM THE TITANIC DATASET")
    print("=" * 55)

    print(f"\nNumber of passengers: {len(df)}")
    print(f"Number of variables: {df.shape[1]}")

    if "Survived" in df.columns:
        print(
            f"Overall survival rate: "
            f"{df['Survived'].mean() * 100:.1f}%"
        )

    if {"Sex", "Survived"}.issubset(df.columns):
        print("\nSurvival rate by sex:")
        print(
            (df.groupby("Sex")["Survived"].mean() * 100)
            .round(1).to_string()
        )

    if {"Pclass", "Survived"}.issubset(df.columns):
        print("\nSurvival rate by passenger class:")
        print(
            (df.groupby("Pclass")["Survived"].mean() * 100)
            .round(1).to_string()
        )

    print("\nMissing values by column:")
    print(
        df.isnull().sum()[
            df.isnull().sum() > 0
        ].to_string()
    )

    print("\nDuplicate rows:", df.duplicated().sum())

    # 13. CONCLUSION
    display(Markdown("""
    ## 11. Conclusion

    The Exploratory Data Analysis of the Titanic dataset examined
    passenger characteristics, missing values, survival rates, age
    and fare distributions, and correlations among numerical variables.

    The analysis helps identify patterns in passenger survival.
    Survival rates differed by sex and passenger class in this dataset.
    Missing values were also identified, indicating areas that may
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require data preparation before predictive modeling.

This project demonstrates the use of Python, Pandas, NumPy,
and Matplotlib for data exploration, statistical summaries,
and visualization.

**Note:** These findings describe associations in the dataset
and do not establish causation.
"""))

print("\nEDA COMPLETED SUCCESSFULLY!")
print("Review the tables, graphs, and findings above.")
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